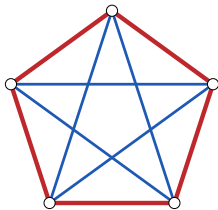


Ramsey Theory

Section 1.8

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



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Today's lecture

1. Classical Ramsey Numbers

**2. Exact Numbers and
Bounds**

3. Graph Ramsey Theory

4. Examples and Exercises

A party problem

Question

At a party of six people, must there be three mutual acquaintances or three mutual strangers?

Model people by vertices of K_6 . Color an edge red for acquaintance and blue for stranger.

We seek a triangle all of whose edges have the same color.

Ramsey numbers

Classical Ramsey number

For positive integers p, q , the Ramsey number $R(p, q)$ is the least n such that every red–blue coloring of the edges of K_n contains a red K_p or a blue K_q .

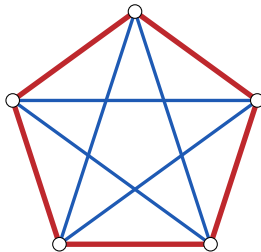
Existence is part of Ramsey's theorem; it is not assumed in the definition.

$$R(p, q) = R(q, p), \quad R(1, q) = 1, \quad R(2, q) = q.$$

Proof

Interchange the colors for symmetry. A single vertex is a copy of K_1 . If there is no red edge in K_q , all edges are blue. Conversely, an all-blue K_{q-1} has neither a red K_2 nor a blue K_q .

Why five people do not suffice



The red edges form a 5-cycle; so do the blue edges.

The lower bound for $R(3, 3)$

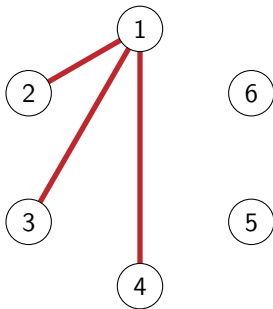
Proof

Neither color class in the displayed coloring of K_5 contains a triangle. Therefore

$$R(3, 3) > 5.$$

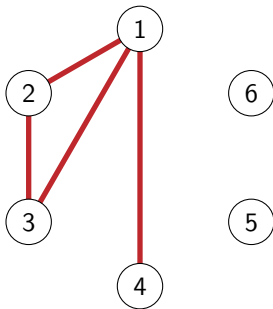
A lower bound requires *one* coloring avoiding both targets. An upper bound must work for *every* coloring.

The upper bound: choose a vertex



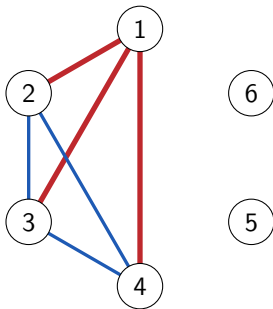
At least three of the five edges at vertex 1 have one color.

The upper bound: first possibility



If one edge among the three neighbors is red, there is a red triangle.

The upper bound: second possibility



If none is red, the three neighbors form a blue triangle.

The first nontrivial Ramsey number

Theorem

$$R(3, 3) = 6.$$

Proof

The coloring of K_5 proves the lower bound 6. The argument at an arbitrary vertex of K_6 proves the upper bound 6.

The fundamental recurrence bound

Theorem

For $p, q \geq 2$,

$$R(p, q) \leq R(p - 1, q) + R(p, q - 1).$$

This inequality proves that all classical Ramsey numbers are finite by induction on $p + q$.

Proof of the recurrence bound

Proof

Write $a = R(p-1, q)$ and $b = R(p, q-1)$, and color K_{a+b} . At a vertex v , let A be its red neighbors and B its blue neighbors.

Since $|A| + |B| = a + b - 1$, either $|A| \geq a$ or $|B| \geq b$.

In the first case, A contains a blue K_q or a red K_{p-1} , which extends with v to a red K_p . The second case follows by interchanging the colors.

A binomial upper bound

Corollary

For $p, q \geq 2$,

$$R(p, q) \leq \binom{p+q-2}{p-1}.$$

Proof

The bound is exact on the boundary $p = 2$ or $q = 2$. Induct on $p + q$ and use Pascal's identity:

$$\binom{p+q-3}{p-2} + \binom{p+q-3}{p-1} = \binom{p+q-2}{p-1}.$$

Using the recurrence

$$R(3, 4) \leq R(2, 4) + R(3, 3) = 4 + 6 = 10.$$

$$R(4, 4) \leq 2R(3, 4) \leq 20.$$

These are upper bounds. The recurrence need not give the exact value.

A parity argument sometimes improves the bound by 1.

The parity refinement

Theorem

If $a = R(p - 1, q)$ and $b = R(p, q - 1)$ are both even, then

$$R(p, q) \leq a + b - 1.$$

Proof

Suppose a coloring of K_{a+b-1} avoids both targets. At every vertex, red degree is at most $a - 1$ and blue degree at most $b - 1$. Their sum is $a + b - 2$, so both bounds are attained.

The red graph has odd order $a + b - 1$ and every degree $a - 1$ is odd, contradicting the handshaking lemma.

The upper bound for $R(3, 4)$

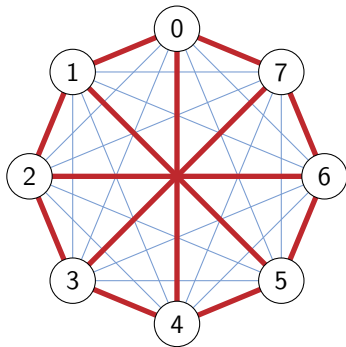
$$R(2, 4) = 4, \quad R(3, 3) = 6.$$

Both numbers are even. Hence

$$R(3, 4) \leq 4 + 6 - 1 = 9.$$

To prove equality, we need a coloring of K_8 with no red triangle and no blue K_4 .

A coloring on eight vertices



Red: an 8-cycle and the four opposite-vertex edges.

Why the coloring works

Proof

The red graph has no triangle: the neighbors of a vertex are its two cyclic neighbors and its opposite vertex, and no two of these are adjacent in red.

A blue K_4 would be an independent set of size 4 in the red graph. An independent 4-set on an 8-cycle must consist of alternate vertices. But either alternating set contains an opposite-vertex red edge.

Thus the coloring avoids both targets, and $R(3, 4) \geq 9$.

An exact value and a consequence

Theorem

$$R(3, 4) = 9.$$

Therefore

$$R(4, 4) \leq R(3, 4) + R(4, 3) = 18.$$

Obtaining a matching lower bound requires a separate construction.

The diagonal case

$$R(k, k) \leq \binom{2k-2}{k-1} < 4^{k-1} \quad (k \geq 2).$$

The proof gives a finite bound, but exact values become difficult quickly.

Question

Can we also show that diagonal Ramsey numbers grow rapidly?

A random-coloring argument

Color each edge of K_n independently red or blue, each with probability $1/2$. For a fixed set of $k \geq 2$ vertices, the probability of a monochromatic clique is

$$2 \left(\frac{1}{2} \right)^{\binom{k}{2}} = 2^{1 - \binom{k}{2}}.$$

There are $\binom{n}{k}$ candidate vertex sets. By the union bound,

$$\Pr(\text{some monochromatic } K_k) \leq \binom{n}{k} 2^{1 - \binom{k}{2}}.$$

A probabilistic lower bound

Theorem

If

$$\binom{n}{k} 2^{1-\binom{k}{2}} < 1,$$

then $R(k, k) > n$.

Proof

With positive probability, the random coloring has no monochromatic K_k . Therefore at least one such coloring exists.

An existence proof need not explicitly display the coloring.

Multicolor Ramsey number

$R(k_1, \dots, k_r)$ is the least n such that every r -coloring of $E(K_n)$ has a K_{k_i} in color i for some i .

The same neighborhood argument proves finiteness, using induction on the sum of the parameters.

The two-color version is the main tool in this section.

Graph Ramsey numbers

Graph Ramsey number

For graphs G, H , let $R(G, H)$ be the least n such that every red–blue coloring of K_n has a red copy of G or a blue copy of H .

Copies are subgraphs, not necessarily induced subgraphs.

$$R(G, H) \leq R(|V(G)|, |V(H)|).$$

An immediate lower bound

Observation

For graphs G, H of orders m, n ,

$$R(G, H) \geq \min\{m, n\}.$$

A stronger construction uses the chromatic number of one graph and the size of a connected component of the other.

A structural lower bound

Theorem

Let $c(G)$ be the largest order of a component of G . Then

$$R(G, H) \geq (c(G) - 1)(\chi(H) - 1) + 1.$$

Proof

Take $\chi(H) - 1$ disjoint red cliques, each of order $c(G) - 1$, and color all edges between them blue. No red component is large enough to contain the largest component of G . The blue graph is $(\chi(H) - 1)$ -partite, so it cannot contain H .

Trees versus complete graphs

Theorem

For a tree T_m on $m \geq 2$ vertices and $n \geq 2$,

$$R(T_m, K_n) = (m - 1)(n - 1) + 1.$$

The structural lower bound gives the right-hand side. We prove the matching upper bound using chromatic number.

A tree embedding lemma

Lemma

Every graph with minimum degree at least $m - 1$ contains every tree of order m .

Proof

Induct on m . Delete a leaf u of the tree, with neighbor v . By induction embed the remaining tree. The image of v has at least $m - 1$ neighbors, but at most $m - 2$ other vertices are already used. An unused neighbor can receive u .

Trees versus complete graphs: upper bound

Proof

Color K_N , where $N = (m-1)(n-1) + 1$, and suppose there is no blue K_n . The red graph F then has independence number at most $n-1$. Thus

$$\chi(F) \geq \left\lceil \frac{N}{n-1} \right\rceil \geq m.$$

Choose a subgraph minimal by vertex deletion with chromatic number at least m . Every vertex has degree at least $m-1$; otherwise an $(m-1)$ -coloring after deletion extends back. The lemma gives a red T_m .

An example of the tree formula

Example

Let P_5 be the path on five vertices. Then

$$R(P_5, K_3) = (5 - 1)(3 - 1) + 1 = 9.$$

For the lower bound, use two red K_4 's with all edges between them blue.

The theorem says every coloring of K_9 has a red P_5 or a blue triangle.

Question

Show that every red–blue coloring of K_{10} has a red tree on four vertices of any prescribed shape, or a blue K_4 .

What construction shows that 9 vertices do not suffice?

$$R(T_4, K_4) = (4 - 1)(4 - 1) + 1 = 10.$$

For K_9 , use three disjoint red triangles and make every edge between triangles blue.

Red components have only three vertices, and a blue clique uses at most one vertex from each triangle.

- Ramsey theory forces ordered substructures in sufficiently large colorings.
- Lower bounds come from constructions; upper bounds apply to every coloring.
- Neighborhood arguments give recurrence and binomial upper bounds.
- Graph Ramsey numbers connect coloring, connectivity, and tree embeddings.

Some Essential Counting Problems